A dissertation on

AUTOMATED FILTRATION OF SYNTHETIC IMAGES USING A LIGHTWEIGHT SIGLIP 2 HYBRID ARCHITECTURE

submitted in partial fulfillment for the award of the degree of

Master of Science in Data Science

by

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April - 2026

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DECLARATION

I hereby declare that the dissertation – II entitled **Automated Filtration of synthetic images using a lightweight SigLIP 2 Hybrid architecture** submitted by me to the Department of Mathematics, School of Advanced Sciences, Vellore Institute of Technology, Chennai, in partial fulfillment of the requirements of the award of the degree of **Master of Science in Data Science** is a bona fide record of the work carried out by me under the supervision of **Dr. Hannah Grace G**.

I further declare that the work reported in this dissertation – II, has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this Institute or of any other Institute or University.

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This is to certify that the dissertation-II entitled "**Automated Filtration of synthetic images using a lightweight SigLIP 2 Hybrid architecture**" is prepared and submitted by **Arun Kumar M (Reg. No. 24MDT1131)** to the Department of Mathematics, School of Advanced Sciences, Vellore Institute of Technology, Chennai, in partial fulfillment of the requirement for the award of the degree of **Master of Science in Data Science** is a bona fide record carried out by him under my supervision, as per the VIT code of academics and research ethics.

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ACKNOWLEDGEMENT

With immense pleasure and a deep sense of gratitude, I wish to express my sincere thanks to my supervisor *Dr. Hannah Grace G*, Associate Professor, School of Advanced Sciences, Vellore Institute of Technology (VIT), Chennai for her motivation and continuous encouragement, without which this project would not have been successfully completed.

I am deeply grateful to the honorable Chancellor of VIT, *Dr. G. Viswanathan*; the Vice Presidents, *Dr. Sankar Viswanathan*, *Dr. Sekar Viswanathan*, and *Dr. G. V. Selvam*; the Executive Director, *Dr. Sandhya Pentareddy*; the Assistant Vice President, *Ms. Kadhambari S. Viswanathan*; the Vice Chancellor, *Dr. V. S. Kanchana Bhaaskaran*; the Pro Vice Chancellor of VIT Chennai, *Dr. T. Thyagarajan*; the Director, Chennai Campus, *Dr. K. Sathiyanarayanan*; and the Additional Registrar, *Dr. P. K. Manoharan*, for motivating me to carry out this dissertation work at Vellore Institute of Technology, Chennai.

I express my sincere thanks to *Dr. Vinitha G*, Dean, School of Advanced Sciences, VIT, Chennai and *Dr. Manivannan A*, Head, Department of Mathematics, School of Advanced Sciences, VIT, Chennai for their support and encouragement.

I wish to extend my profound sense of gratitude to *my parents* for providing me the moral support and encouragement whenever required.

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ABSTRACT

Generative AI is exploding in popularity, and it's really changed how we think about whether digital media is real. The image makers we have now, using diffusion, Generative Adversarial Networks or GANs, and systems that work with multiple kinds of data to create images, can make photos that look absolutely real to people. This makes it easy for false information, scams, and a decline in faith in anything we see online to spread around. As for checking if an image is real, the tools we have at the moment - older Convolutional Neural Networks, and the more demanding Vision-Language Models - become much less accurate with pictures that have a lot going on, are in a particular artistic style, or have words on them. The industry has largely compensated for this deficiency through multi-model ensemble architectures, however, these systems are computationally prohibitive, resource-intensive, and fundamentally too slow for real-time web deployment. This dissertation presents a novel, lightweight, single-pass hybrid AI image detector that matches the robustness of heavy ensembles while operating at a fraction of the latency. The proposed architecture isolates the spatial feature extraction capabilities of the Sigmoid Loss for Language-Image Pre-Training (SigLIP) 2 vision encoder by deliberately removing its text-processing components, thereby reducing memory footprint and inference cost. To capture fine-grained forensic details invisible at standard resolutions, the model's position embeddings are bicubically interpolated from 224×224 to 512×512. These rich deep embeddings are fused with seven explicitly computed traditional frequency and statistical forensic signals—Local Binary Patterns (LBP), Discrete Cosine Transform (DCT), Gradient Co-occurrence, Haar Wavelet Energy, Fast Fourier Transform (FFT) Power Slope, Bayer Noise Pattern, and Edge Sharpness Grid—which collectively address the blind spots of deep learning alone. A custom three-layer classification head employing Squeeze-and-Excitation (SE) style channel attention is then applied to the fused embedding to identify AI-generated artefact patterns. Evaluated on a stratified 15% held-out test split of the training dataset (1,132

images), the model achieves 96.6% accuracy, 98.9% precision, 95.3% recall, 97.0% F1-score, and a 0.992 AUC-ROC, with an inference latency of 142.4 milliseconds measured on a dual NVIDIA Tesla T4 GPU environment on Kaggle Notebook. A comparative analysis against commercial detection platforms Undetectable AI (5,230 ms) and AI or Not (920 ms) demonstrates that our architecture correctly identifies hard-negative AI-generated images misclassified as real by both commercial tools, while achieving a $6.5\times$ and $36.7\times$ speed advantage respectively. The model is deployed as an interactive Hugging Face Spaces application, providing accessible, real-time image verification.
Keywords: AI Image Detection, Synthetic Media, Vision-Language Models, SigLIP 2, Feature Fusion, Image Authentication.

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List of Symbols, Abbreviations and Nomenclature

AI	:	Artificial Intelligence
AUC-ROC	:	Area Under the Receiver Operating Characteristic Curve
BN	:	Batch Normalisation
CNN	:	Convolutional Neural Network
DCT	:	Discrete Cosine Transform
DWT	:	Discrete Wavelet Transform
FFT	:	Fast Fourier Transform
FNR	:	False Negative Rate
FPR	:	False Positive Rate
GAN	:	Generative Adversarial Network
GELU	:	Gaussian Error Linear Unit
GPU	:	Graphics Processing Unit
GRU	:	Gated Recurrent Unit
LBP	:	Local Binary Pattern
MLP	:	Multi-Layer Perceptron
NCII	:	Non-Consensual Intimate Imagery
SE	:	Squeeze-and-Excitation
SigLIP	:	Sigmoid Loss for Language-Image Pre-Training
TTS	:	Text-to-Speech
VLM	:	Vision-Language Model
ViT	:	Vision Transformer
VRAM	:	Video Random Access Memory

Chapter 1

Introduction

1.1 Background

In the current digital era, an unprecedented level of convergence has been witnessed with regard to computational power and generative intelligence, leading to an irreversible transformation with regard to the nature of visual content. Over the course of a few years, diffusion models, Generative Adversarial Networks, and multimodal language-image systems have progressed from generating basic approximations of human faces to producing high-resolution photorealistic images that are indistinguishable from photographs taken with high-quality equipment. Today, DALL·E (OpenAI), Stable Diffusion (via Leonardo AI), Google Gemini, Ideogram, and OpenAI Sora collectively produce millions of synthetic images on a daily basis, with no standardized method for tracking the diffusion of these images on social media feeds, news aggregators, and messaging platforms. Such an eventuality has significant implications for society. Synthetic images have been linked to the creation of non-consensual intimate imagery, political actions that are falsified, fraudulent financial statements, and the degradation of the integrity of evidence in legal cases. An analysis performed by the RAND Corporation [1] indicates that the adverse effects of AI-generated imagery are part of a spectrum of harm that includes personal safety threats and geopolitically focused disinformation. Notably, the adverse effects cannot be addressed with human vigilance, with studies indicating that even trained individuals have performance levels near chance when distinguishing photographs and AI-generated imagery.

Automated detection systems, therefore, represent the essential last line of defense in the integrity of digital content. The domain of image detection from AI-generated content, however, is highly fragmented and in constant evolution. Initial detection tools, based on convolutional neural networks (CNN) and designed to recognize specific types of GANs, such as ProGAN or StyleGAN, are found to have very low generalizability to the newer diffusion-based GANs. More complex networks, based on Vision Transformers or Vision-Language Models, while more robust, are associated with computational costs that are prohibitively high, rendering them impractical in the context of real-world, web-scale, timely detection. In practice, the solution has been the development of ensemble detection tools, which, while more accurate, are associated with costs that are only accessible to large corporations with access to GPU computing infrastructure.

1.2 Problem Statement

The primary research question that is being addressed by the dissertation is whether the same level of precision and robustness, as is achieved by the computationally expensive and complex multi-model ensemble-based architectures, can be achieved by the single-pass artificial intelligence image detector, while maintaining the speed and resource requirements that are appropriate and feasible within the web-based application domain. The state-of-the-art image detectors have shown that, while high precision is achieved, the speed and resource requirements are high, and while the speed and resource requirements are achieved, the precision is very low and hence not feasible. So far, no prior work exists on the application and implementation of the vision encoder part of the Vision-Language Model, which was specifically pre-trained using the pairwise sigmoid loss function, called SigLIP 2, as the back end for the image forensics application using artificial intelligence. There is no prior implementation that combines the traditional image forensic features with the back end using a single classification head.

1.3 Motivation

This research is inspired both by technical and social factors. The social concern is that the digital visual, like the image, loses its credibility. It's a main danger to democratic discourse, judicial procedure, and personal well-being. It is crucial that journalists, fact-checkers, legal experts, and ordinary citizens have access to fast, accurate and freely accessible detecting tools to navigate an information environment saturated with AI-generated content.

Technically, the swift pace of change in generative models has consistently outpaced the development of effective detection countermeasures. The most recent benchmark work by Li et al. [2] revealed that even modern state-of-the-art detectors exhibit drastic performance decline under realistic conditions such as JPEG compression, image resizing, and colour changes. Ren et al. [3] further demonstrated that no single existing architecture performs consistently across all benchmarks, and that training data alignment is a key performance determinant over architectural sophistication. These results indicate that architectures combining deep learned representations with explicit, physics-grounded forensic signals could provide a more resilient and general detection system than purely data-driven systems.

1.4 Challenges

Several significant challenges characterise the problem of automated AI image filtration.

Generalization across generative paradigms: Models trained on GAN generated Images frequently fail on diffusion model outputs, as the two paradigms leave fundamentally different artifact signatures. A detector must be robust across both.

Hard negatives: AI generators are increasingly designed with photorealism as an explicit optimization target. Certain images are particularly high-resolution outputs from models like Stable Diffusion 3 or DALL-E 3 exhibit no obvious visual artifacts and can defeat conventional detectors.

Computational constraints: For real-time web deployment, inference latency are required to be less than 200 milliseconds on conventional hardware. Current ensemble methods go far beyond this limit.

Dataset bias: The public training sets tend to be biased in favour of certain generative models due to the popularity of trendy models, being biased in the opposite direction in relation to the diversity of photography in the real world.

Resolution mismatch: The traditional vision models are based on a resolution of 224×224 pixels, which in the case of fine-grained forensic artifacts, particularly those involving frequency statistics, results in the degradation of artifacts due to downsampling.

Adversarial resilience: Post-processing operations on images, including JPEG compression, resizing, and noise addition are intended to mask artifact signatures, which are particularly targeted in frequency domain-based artifact detection.

1.5 Essence of the Approach

This dissertation proposes a hybrid forensic architecture that addresses the above challenges through 3 principal innovations. First, the text component of the SigLIP 2 model [4] is removed, leaving a vision-only encoder that has been enriched through image-text pretraining, yielding richer, more semantically dense visual representations than a standard image classification model. The position embeddings of this encoder are bicubically interpolated from the standard 224×224 training resolution to 512×512, preserving fine-grained forensic details that would otherwise be lost.

Second, seven traditional forensic signals derived from established signal processing and statistical principles are LBP entropy, DCT Benford law deviation, gradient co-occurrence entropy, Haar wavelet diagonal energy, FFT power slope deviation, Bayer noise inter-channel correlation, and edge sharpness distribution are explicitly computed from the raw image pixels and embedded by a dedicated Multi-Layer Perceptron (MLP). These features capture physical properties of real camera images that deep learning models are known to overlook.

Third, the deep embedding from the SigLIP 2 backbone and the forensic feature embedding are concatenated and passed through a custom three-layer classification head employing SE-style channel attention, which learns to weight the contribution of each feature dimension based on its relevance to the AI/real discrimination task.

1.6 Statement of Assumptions

The proposed system operates under the following assumptions:

- The input image is a standard RGB digital image in a common format (JPEG, PNG, or WebP).
- The detector is used as a binary classifier: outputs are *AI-generated* or *Real*, with no distinction between generative sub-types.
- The system is evaluated under an open-world detection scenario where the specific generative model that produced an AI image is unknown at inference time.
- Hardware availability: The training ground used dual NVIDIA Tesla T4 GPUs which are available via the Kaggle Notebooks environment. On the other hand, the inference environment will be performed in CPU hardware on Hugging Face Space.
- The dataset is limited however (curated from different data sources) and we can't expect to generalize to every new generative model which may be released after the delivery date of our dataset.

1.7 Aims and Objectives

Aim: The aim of this research is to design, implement, and evaluate a lightweight, single-pass hybrid AI image detector that matches the accuracy of heavy ensemble models while being fast enough for real-time web deployment.

Objectives:

1. Create a heterogeneous and diverse collection of the base 7,600 images across various real-world photo sources and five and more generative AI paradigms.

2. Customize the SigLIP 2 vision encoder on our forensic image classes by removing the text bits and bicubically interpolating the position place to 512 x 512 resolution.

3. Develop and validate a set of seven classical forensic feature extractors encompassing texture (LBP), frequency (DCT, FFT, Haar), noise (Bayer), and edge structure (Gradient Co-occurrence, Edge Sharpness), and test their discriminative power individuallyBuild and benchmark a set of seven classical forensic feature extractors for the forensic domain spanning texture (LBP), frequency (DCT, FFT, Haar), noise (Bayer), and edge structure (Gradient Co-occurrence, Edge Sharpness) and benchmark their discriminative capability in isolation.

4. Train a classically hybrid classification architecture that combines deep SigLIP 2 features with the classical forensic feature vector via a dedicated SE-attention head.

5. Benchmark our hybrid model on our testing data (as above) with the aim of greater than 95% accuracy and greater than 0.90 AUC-ROC.

6. Benchmark our deployed model against commercial detection solutions (Undetectable AI, AI or Not) on our hard-negative images, with a slight nod towards accuracy and clear wins on speed.

7. Deploy our trained model to Hugging Face Spaces as an interactive web app and deliver an open and operationally accessible image verification tool.

1.8 Organisation of the Report

The remainder of this dissertation is organised as follows. Chapter 2 is an overview of the literature on deep learning, Vision Transformers, Vision-Language models, and detection of AI-generated images. Chapter 3 reflects further on the research problem and states the particular gap in knowledge we aim to fill. Chapter 4 details the entirety of our methodology, including construction of our dataset, design of forensic features, the model itself, training strategy and deployment. Chapter 5 contains description of experimental results, ablations, ROC curves analysed, and comparison against a commercial baseline. Finally Chapter 6 discusses our findings and their limits, as well as future work we intend to explore.

Chapter 2

Literature Review

2.1 Deep Learning and Vision Transformers

Deep learning has revolutionized almost every aspect of artificial intelligence, with groundbreaking success in practically all of computer vision and natural language processing. Goodfellow et al. [5] give a maths heavy account of neural networks, including optimization methods, regularization techniques, and architectural 'hierarchies' which give deep networks their representational power. For image tasks, deep residual networks were introduced to combat the vanishing gradient problem using identity shortcut connections [6], enabling extremely deep networks to be trained and achieve record results on huge benchmarks such as ImageNet.

The Vision Transformer (ViT) shifted attention from CNNs to attention based models by demonstrating that a pure transformer architecture over a sequence of image patches can be competitive with convolutional architectures if sufficiently large [7]. Unlike CNNs, which implicitly encode inductive biases of spatial locality and translation equivariance into their architectures, ViT models are required to learn from scratch to attend to spatial aspects of an image through self-attention, providing them with the capacity to capture long-range dependencies of particular use in forensic tasks for which artifact signatures may lie in non-local parts of the image.

A systematic layer-wise investigation of CLIP-based Vision Transformers for AI-generated image detection by Park et al. [8] shows that intermediate layers of ViT models rather than the final classification layer provide the most generalizable forensic discrimination features. They propose a Mixture of Layers for Detection (MoLD) mechanism that dynamically combines features across several ViT layers through a learned gating mechanism, yielding better generalization to previously unseen generative models. This observation supports the theory that intermediate feature representations acquired during vision-language pre-training encode forensically meaningful characteristics that are lost in last-layer compression.

2.2 Vision-Language Models and SigLIP 2

The intersection of visual and linguistic representation learning has led to a family of capable foundation models with exceptional cross-modal understanding. CLIP [9] established the format of contrastive vision-language pre-training, aligning image and text representations through a contrastive objective that pulls similar image-text pairs to share similar embeddings. Trained on 400 million image-text pairs, the resulting model demonstrated strong zero-shot transfer to image classification tasks.

Liu et al. [10] proposed FatFormer, a forgery-aware adaptive transformer that explicitly utilises the vision-language spaces of pre-trained CLIP for generalised synthetic image detection. FatFormer combines local forgery traces in both image and frequency domains using a forgery-aware adapter module, and uses language-guided alignment to regularise the learned features. The approach scored 98% average accuracy on unseen GAN architectures and 95% on unseen diffusion models, demonstrating the value of vision-language pre-training as a foundation for forensic detection.

SigLIP 2, proposed by Tschannen et al. [4], is a significant improvement to CLIP-based architectures. Rather than using a contrastive loss with softmax normalisation that requires large batch sizes for effective training, SigLIP 2 adopts a pairwise sigmoid loss where each image-text pair is trained independently. The model was trained on the WebLI dataset comprising 10 billion image-text pairs and supports distortion-free native image resolutions through a flexible resolution handling system named NaFlex. These properties combine to form a vision encoder with highly dense and semantically rich feature representations—characteristics that make it well suited to the fine-grained discrimination required in image forensics.

2.3 AI-Generated Image Detection: State of the Art

The challenge of distinguishing AI-generated from real images has evolved significantly since early GAN detection work. Bonettini and Bestagini [11] capitalised on the statistical characteristics of Benford's Law in the DCT coefficient distributions of GAN-generated images, observing that the natural distribution of significant digits of real camera images is not replicated by GAN synthesis. This frequency-domain perspective provided a significant foundation for the forensic feature engineering approach used in the present work.

Diffusion model-based generators rendered the detection landscape considerably more difficult. Li et al. [2] proposed AIGIBench, a large benchmark comprising 23 diverse subsets of AI-generated images from a variety of generative models, compared against real images from FFHQ, CelebA-HQ, and Open Images V7. Their evaluation of 11 state-of-the-art detectors demonstrated that all current methods exhibit steep accuracy deterioration under real-world conditions including JPEG compression, image resizing, and colour jitter, with some detectors declining to near-random performance under modest perturbation.

Ren et al. [3] extended these findings with the first large-scale evaluation of 16 detection approaches in a zero-shot setting across 12 diverse datasets, including GenImage, WildFake, OpenFake, and SynthBuster. Their results revealed enormous performance variation across datasets, with the model performing best on one benchmark potentially performing worst on another, and proposed that training data compatibility is a more important performance predictor than algorithmic design choices. These results encourage the creation of more eclectic training datasets and more abstract feature representations.

Perotti et al. [12] demonstrated that available open-source deepfake detectors are not capable of generalising across different generative models. Through extensive experimentation, they demonstrated that ensemble methods with latent feature fusion are reliable for improving single binary classifiers. However, the authors discussed the fact that ensemble advantages are acquired at high computational cost and thus efficient single model alternatives are important.

Karageorgiou et al. [13] created SPAI (Spectral AI generated Image detection), an approach based on masked spectral learning to identify AI images as out-of-distribution samples by determining their similarity to a spectral reconstruction distribution that has been trained on real images. Evaluated on generative models not seen during training, SPAI improved accuracy over state-of-the-art methods by 5.5%, demonstrating the strength of spectral domain analysis for generalized detection.

Zhong et al. [14] proposed PatchCraft, a preprocessing scheme that exploits differences in inter-pixel correlation between rich- and poor-texture regions of AI-generated images. PatchCraft disrupts global semantic coherence by shuffling texture patches, forcing a detector to attend to local texture statistics, achieving an average zero-shot accuracy of 89.85% across six generative models.

2.4 Hybrid Architectures for Efficient Detection

Beyond pure approaches of deep learning, a range of studies have been conducted on hybrid solutions which mix the power of deep learning with the interpretability and hand-crafted features. Lokner Ladi’c et al. [15] proposed a model which jointly uses a CNN along with a wavelet entropy technique, achieving 97.32% accuracy whilst remaining computationally acceptable due to being simpler. Their explainability analysis showed that in this architecture the model focuses primarily on certain complex textures of the kind found in backgrounds, which is very similar to the LBP and gradient co-occurrence forensic signals used in our work.

Gupta and McMahan [16] showed that preprocessing the input images fed to a CNN based architecture with FFT leads to +3.27% improved accuracy and 10.06% reduced loss compared to models that do not use frequency-domain features, justifying our use of the FFT based power slope forehead signal in the present architecture.

Rani et al. [17] present a model using VisionTransformers and Linformer structure developed to detect deepfake videos with an accuracy of 98.9% whilst being more computationally efficient than the standard ViT structures. Harshit et al. [18] have in fact presented more recently a model also using EfficientNet-B0, along with a GRU, used to detect deepfake videos.

2.5 Traditional Forensic Signals in the Deep Learning Era

A growing area of research is focused on the exploration of different approaches to integrate the available traditional data features into deep learning models. It is a known fact that modern deep learning architectures are plagued with a number of limitations. Tan et al. [19] showed that gradient maps are a universal representation for detecting GAN-generated images, improving performance by 11.4

Significant research has been conducted in photo forensics, studying camera sensor noise as a forensic signal. Photographs, during Bayer demosaicing, have inherent correlations between color channel noise residuals, which are absent in AI-generated images, since color values in all three channels are synthesized simultaneously. Hence, a discriminative signal, robust against post-processing operations affecting the physical content of images, has been obtained.

The Local Binary Pattern (LBP) was originally proposed by Ojala et al. [20] for texture classification, which has been extended significantly for image authenticity verification. The use of Shannon entropy values for LBP has been shown to be effective, since a scalar value representing statistical distribution of micro-textures in images has been obtained. Photographs, being an inhomogeneous mixture of edges, gradients, sensor noise, and physical surface textures, tend to have LBP entropy values closer to the theoretical limit, while AI-generated images tend to have LBP values that systematically deviate from those obtained from real photographs, since AI-generated images, which aim to be coherent at a macroscopic level, tend to be poor in physical realism at a micro-level. Hence, LBP has been proposed as a discriminative feature, distinguishing between AI-generated images and real photographs, with an area under the curve (AUC) value of 0.79 during the feature selection phase of this research, making LBP the best discriminative feature among traditionally used features.

2.6 Literature Gap and Research Positioning

A comprehensive review of the existing literature reveals a critical gap at the intersection of three research directions: the use of VLM-pretrained vision encoders with a sigmoid pairwise loss (SigLIP 2), the explicit combination of deep embeddings with traditional forensic signals in a unified hybrid architecture, and the rigorous evaluation of such an architecture against hard-negative images and commercial detection baselines under realistic deployment constraints.

While FatFormer [10] demonstrated the value of CLIP-based vision-language pre-training for detection, it does not explore the SigLIP 2 encoder, does not incorporate traditional forensic signals, and does not report inference latency under web-deployment conditions. While Gupta and McMahan [16] and Tan et al. [19] demonstrated the value of frequency-domain and gradient features respectively, neither work combined these signals with a VLM-quality vision backbone. The benchmark studies of Li et al. [2]

and Ren et al. [3] establish the severity of the generalisation problem but offer no architectural solutions. The present dissertation directly addresses this gap by proposing, implementing, and rigorously evaluating a hybrid architecture that, to the best of the author's knowledge, is the first to combine a SigLIP 2 vision encoder with an explicit traditional forensic feature branch for AI image detection.

Chapter 3

Research Problem: Bridging Deep Visual Forensics and Traditional Signal Analysis

3.1 Formal Problem Definition

Let $\mathcal{I}$ denote the set of all digital images. The binary classification problem is defined as: given an image $x \in \mathcal{I}$, determine whether x was produced by a real-world camera or by a generative AI system. Formally, we seek to learn a function $f : \mathcal{I} \to \{0, 1\}$, where 0 denotes AI-generated and 1 denotes real/photographed, such that f minimises the expected binary cross-entropy loss over the joint distribution of real and AI-generated images encountered in practice:

$$\mathcal{L}(f) = -\mathbb{E}_{(x,y)\sim P}\left[y \log f(x) + (1 - y) \log(1 - f(x))\right] \tag{3.1}$$

where P is the distribution over real-world image pairs (x, y).

The challenge inherent in learning f is that the distributions of real and AI-generated images are both high-dimensional and non-stationary: new generative models are continuously released, and the statistical properties of AI-generated images evolve rapidly. A system trained on the outputs of specific generative models at a specific point in time may not generalise to the outputs of models released after training. This motivates the design of f to incorporate signals grounded in physical principles—such as camera sensor physics and natural image statistics—rather than relying exclusively on learned artefact signatures specific to particular generative models.

3.2 Identified Limitations of Existing Approaches

Through analysis of the literature and hands-on evaluation of commercially available detection tools, four principal limitations of existing approaches were identified.

First, single-model CNN and ViT detectors achieve acceptable accuracy on in-distribution test sets—images from the same generative models present in the training data—but exhibit substantial accuracy degradation on out-of-distribution images from unseen generators. This generalisation failure is particularly acute for newer diffusion model architectures, whose synthesis process leaves different artefact patterns than the GAN architectures that dominate most existing training datasets.

Second, the ensemble architectures deployed by commercial detection platforms (represented in this study by Undetectable AI and AI or Not) achieve better generalisation but at prohibitive computational cost. Measured latencies of 5,230 ms and 920 ms respectively for single-image inference render these systems unsuitable for integration into content moderation pipelines that must process thousands of images per second.

Third, existing detection approaches based exclusively on deep learning features have systematic blind spots in the frequency and noise domains. Neural networks trained with stochastic gradient descent on standard image datasets do not develop representations specifically sensitive to the statistical properties of natural image noise, Bayer demosaicing artefacts, or the log-log frequency slope of natural image power spectra. Explicitly engineering features that capture these properties and fusing them with deep embeddings can address these blind spots without requiring a fundamentally different training paradigm.

Fourth, existing architectures typically operate at 224×224 pixel resolution—the canonical input size of ImageNet-pretrained models. At this resolution, fine-grained forensic signals, particularly noise patterns, texture micro-statistics, and high-frequency DCT irregularities, are destroyed by the downsampling process. Operating at 512×512 resolution preserves these signals but requires architectural modifications to accommodate the larger positional embedding space.

3.3 Research Hypotheses

The research is guided by three primary hypotheses:

- **H1:** A vision encoder pre-trained with a pairwise sigmoid loss on 10 billion image-text pairs (SigLIP 2) will provide richer forensic feature representations than architectures pre-trained with standard classification objectives, due to its training to align diverse visual concepts with linguistic descriptions.
- **H2:** Explicitly fusing seven handcrafted forensic features derived from physics-grounded principles of natural image statistics with the SigLIP 2 deep embedding will improve detection accuracy and generalisation compared to using either signal type in isolation.
- **H3:** A hybrid single-model architecture incorporating both signal types can achieve accuracy and generalisation competitive with multi-model ensembles while maintaining inference latency below 200 milliseconds on accessible hardware.

Chapter 4

Methodology

4.1 Overview of the Development Pipeline

The development of the proposed hybrid AI image detector proceeded through four major phases: (i) dataset construction and quality assurance; (ii) forensic feature engineering and Phase-1 feature selection; (iii) model architecture design, training, and hyperparameter optimisation; and (iv) inference pipeline development and web deployment. These phases are described in detail in the following sections. Figure 4.1 illustrates the overall development pipeline.

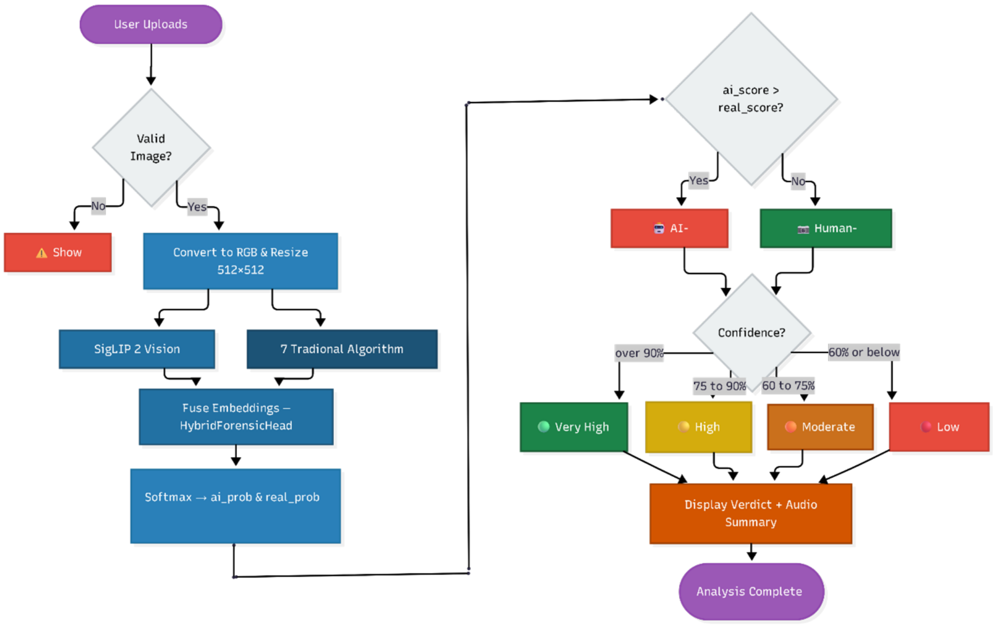

Figure 4.1: Overall development pipeline for the hybrid AI image detector.

4.2 Dataset Construction

The dataset includes photos and AI-generated images from several platforms. This selection matches the variety found in real-world image distributions. We describe the collection methods and the partitioning scheme in the following sections.

4.2.1 Data Collection Strategy

The training data set was designed to reflect the nature and distribution of images that would be commonly found in real-world social media and internet settings, rather than the more typical academic benchmark data sets that have been employed in previous research. The data set was comprised of real-world images, totalling 4,400, gathered from various photography communities, including Reddit and Discord forums, and meticulously curated collections of high-quality images.

Images synthesized using AI, totalling 3,200, were gathered from the image galleries and export tools of five unique AI generative tools: Sora, an Open AI video/image model, Ideogram Explorer, Google Gemini image generation, ChatGPT DALL-E, and Leonardo AI, which employs variants of Stable Diffusion. The above approach was taken in order to ensure that the detector would be trained on images from both major paradigms of image generation: diffusion and multimodal generation, and to avoid overfitting against particular artifact patterns associated with a particular tool or tools.

4.2.2 Dataset Splits

The dataset consists of 7,600 images. These images are divided into 3 partitions called training, validation, and test. To divide this into 3 partitions, we used stratified sampling to preserve the class distribution across all splits. The 3 parts are split into 70% for training (5,320 images), 15% for validation (1,140 images), and 15% for testing (1,132 images). Stratification preserved the original 58% real and 42% AI image ratio in each partition. The test set was held out from all model development decisions, including hyperparameter selection, and was used only for final performance reporting.

Table 4.1: Dataset distribution across training, validation, and test splits.

Split	**Total Images**	**Real**	**AI-Generated**
Training	5,320	3,080	2,240
Validation	1,140	660	480
Test (Held-out)	1,132	472	660
Total	7,592	4,212	3,380

4.3 Forensic Feature Engineering

A two-phase feature engineering process was adopted. In Phase 1, a broad set of candidate forensic signals was evaluated individually to identify those with meaningful discriminative power. In Phase 2, the seven selected features were integrated into the hybrid model as a jointly trained forensic branch. The following subsections describe the selection methodology and each individual feature extractor.

4.3.1 Feature Selection Methodology

Before the comprehensive model training process (Phase 2), a feature selection process (Phase 1) was carried out. A comprehensive set of candidate forensic features was extracted and evaluated over the entire set of 7,600 images using the Area Under the Receiver Operating Characteristic Curve (AUC-ROC) as the primary measure of discriminative power. Each feature was tested individually using a simple logistic regression classifier trained on that feature in isolation. Seven features with individual AUC of at least 0.61 were selected for the hybrid model training process. Table 4.2 lists the selected features and their Phase-1 AUC scores.

Table 4.2: Selected forensic features with Phase-1 individual AUC scores.

No.	Feature Name	What It Measures	AUC	Why It Helps
1	LBP Entropy	Texture randomness across pixels	0.79	AI images have unnaturally uniform or over-random texture patterns
2	DCT Blocking	Frequency regularity in 8×8 blocks	0.69	AI images violate natural frequency distributions (Benford's Law)
3	Gradient Co-occ.	Edge structure and complexity	0.68	AI images have smoother, less varied gradient transitions
4	Haar Wavelet Energy	Diagonal detail energy at multi-scale	0.66	AI images lack fine diagonal detail compared to real photographs
5	FFT Power Slope	High-to-low frequency energy ratio	0.64	AI images cluster around -2.5 slope with unusual consistency
6	Bayer Noise Pattern	Camera sensor inter-channel correlation	0.63	Real cameras leave unique noise fingerprints absent in AI images
7	Edge Sharpness Grid	Sharpness across 9 spatial zones	0.61	AI images show unnatural centre-vs-periphery sharpness ratios

4.3.2 Local Binary Pattern (LBP) Entropy

The Local Binary Pattern (LBP) operator characterises the micro-texture structure of an image by comparing each pixel's intensity with its eight immediate neighbours. For each pixel, a binary code is generated: 1 if a neighbour's intensity is greater than or equal to

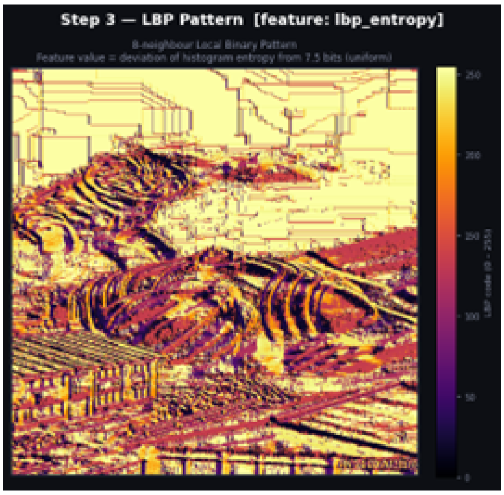

Figure 4.2: LBP entropy feature map visualisation on an AI-generated image.

the middle pixel, or 0 if not. The image is then processed into a 256-bin histogram, and the Shannon entropy of this histogram is used as the forensic signal:

$$H = -\sum_{i=0}^{255} p_i \log_2 p_i \tag{4.1}$$

Entropy reaches a theoretical maximum of about 7.5 bits when the distribution is perfectly uniform across all 256 patterns. This value is very close to what real photographs have, which have a mix of sharp edges, smooth gradients, and fine sensor noise. AI-generated images, which optimize for global aesthetic coherence instead of micro-texture physical realism, create LBP distributions that are consistently different. The feature score is given by the absolute difference from this maximum value scaled to $[0, 1]$ via a sigmoid.

4.3.3 DCT Blocking (Benford's Law Deviation)

The Discrete Cosine Transform (DCT) splits an image into 8×8 pixel blocks. It then turns these blocks into cosine frequency parts. Because cameras use the JPEG standard to handle photos this way, leaving a characteristic statistical fingerprint in the AC coefficients. In particular, the leading digits of these coefficients in real photographs follow Benford's Law, which is the empirical regularity that smaller leading digits

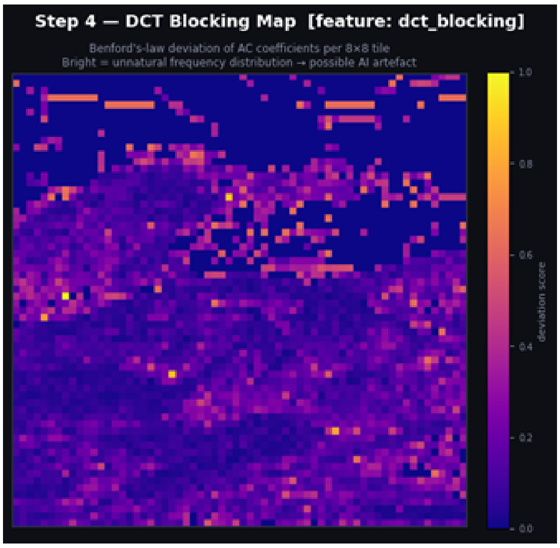

Figure 4.3: DCT Benford's Law deviation visualisation on an AI-generated image.

(1, 2, 3) are more common in natural numerical data than larger ones (7, 8, 9). AI-generated images, synthesised by neural networks that do not replicate the physical optics and compression pipeline of a real camera, produce DCT coefficient distributions that deviate from Benford's Law in measurable ways. The feature is computed as the chi-square deviation between the observed first-digit distribution and the Benford prediction, estimated from up to 400 randomly sampled 8×8 tiles.

4.3.4 Gradient Co-occurrence Entropy

Edges in real photographs arise from physical boundaries between surfaces, changes in illumination, and depth transitions—processes that produce statistically complex, difficult-to-predict sequences of edge strengths. AI generators tend to produce edges that are either overly smooth and uniform or exhibit unnatural repetitive patterns. The gradient co-occurrence feature captures this by computing the gradient magnitude at every pixel, quantising these magnitudes into 32 bins, and building a 32×32 co-occurrence matrix that records the frequency of every pair of consecutive horizontal gradient magnitudes. The Shannon entropy of this co-occurrence matrix is used as the feature signal.

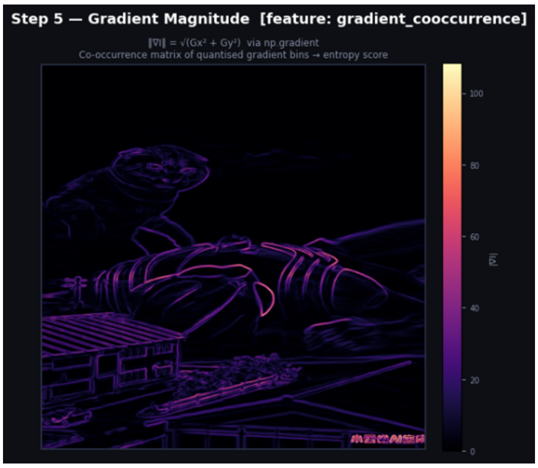

Figure 4.4: Gradient co-occurrence entropy feature visualisation on an AI-generated image.

4.3.5 Haar Wavelet Energy

The Haar Wavelet Transform decomposes an image into four sub-bands at each pyramid level: a low-frequency approximation (LL), horizontal detail (LH), vertical detail (HL), and diagonal detail (HH). The HH sub-band captures edges and fine details oriented diagonally—a pattern that occurs naturally in real-world surfaces, camera optics, and noise, but is systematically suppressed in AI-generated images, which tend to produce locally over-smooth synthetic textures. The feature is computed as the ratio of HH sub-band energy to total detail energy (HH + LH + HL), averaged across three levels of a wavelet pyramid:

$$E_{HH} = \frac{1}{3} \sum_{l=1}^{3} \frac{\|W_{HH}^{(l)}\|^2}{\|W_{HH}^{(l)}\|^2 + \|W_{LH}^{(l)}\|^2 + \|W_{HL}^{(l)}\|^2} \tag{4.2}$$

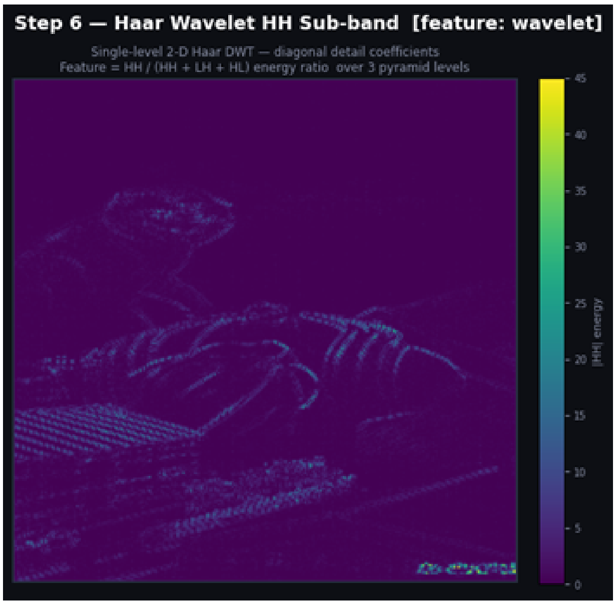

Figure 4.5: Haar wavelet diagonal sub-band energy visualisation on an AI-generated image.

4.3.6 FFT Power Slope

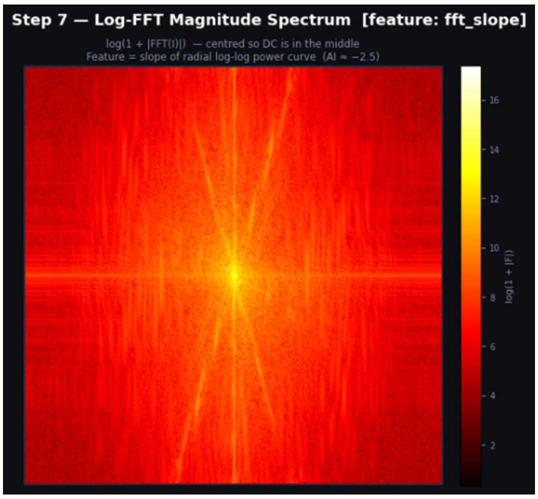

Figure 4.6: FFT power spectrum slope visualisation on an AI-generated image.

The Fast Fourier Transform of a natural photograph, when analysed in log-log frequency space, follows a power-law relationship characterised by a slope of approximately

-2 to -3: high-frequency components (fine details) are consistently less energetic than low-frequency components (large-scale structures). AI-generated images, particularly those from diffusion models and GANs, cluster unusually tightly around a slope of -2.5 with less variance than natural photographs, and sometimes show systematic deviations due to the frequency biases introduced by their decoder architectures. The feature is computed as:

$$s = \left|\hat{\beta} - (-2.5)\right| \tag{4.3}$$

where $\hat{\beta}$ is the fitted radial log-log slope, normalised and passed through a sigmoid function.

4.3.7 Bayer Noise Pattern

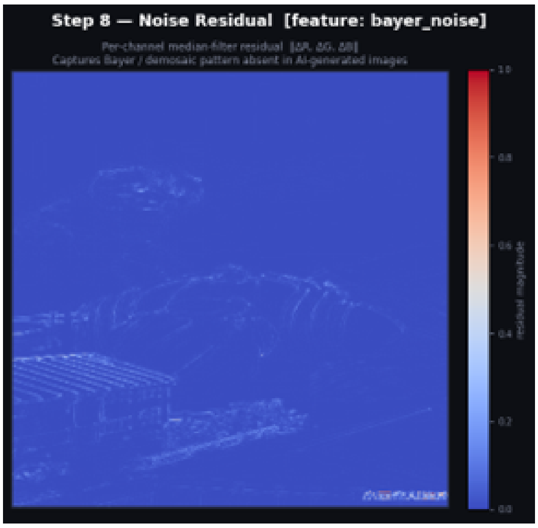

Figure 4.7: Bayer noise inter-channel correlation visualisation on an AI-generated image.

Every digital camera sensor uses a Bayer colour filter array—a mosaic of alternating red, green, and blue photosensors. Because each physical pixel captures only one colour channel, a demosaicing algorithm must interpolate the other two channels from neighbouring pixels. This interpolation process introduces characteristic correlations between the red, green, and blue noise channels that serve as a unique physical fingerprint of real camera hardware. AI generators synthesise all three colour channels simultaneously and directly from learned distributions, so these inter-channel noise correlations are absent. The feature is extracted by applying a 3×3 median filter to each colour channel independently (to isolate the noise residual) and computing a weighted combination of the cross-channel spatial autocorrelation and cross-correlation metrics.

4.3.8 Edge Sharpness Grid

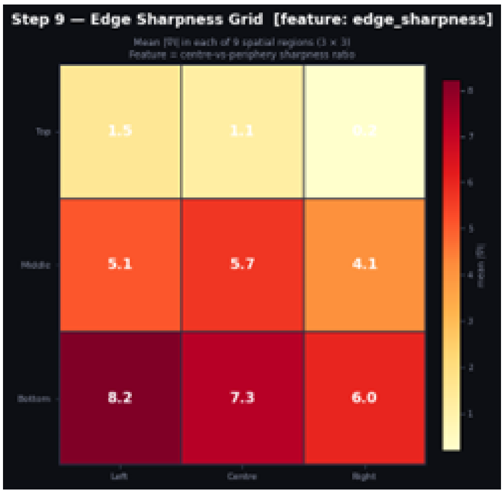

Figure 4.8: Edge sharpness grid visualisation on an AI-generated image.

The optical characteristics of a camera lens—including depth of field, lens sharpness fall-off, and bokeh—create a characteristic spatial distribution of sharpness across a photograph. AI generators tend to produce images with either uniform sharpness across the entire frame, or synthetic depth-of-field effects that do not match the optical physics of any real lens. The feature divides the image into a 3×3 spatial grid of nine regions and measures the mean gradient magnitude (a proxy for local sharpness) in each region, producing a 9-dimensional sharpness distribution vector summarised into a scalar score reflecting the centre-vs-periphery sharpness ratio.

4.4 Model Architecture

The proposed architecture consists of three interconnected components: a SigLIP 2 vision backbone adapted for forensic classification, a dedicated MLP that embeds the traditional forensic feature vector, and a hybrid classification head that fuses both signal streams. Each component is described in detail below.

4.4.1 SigLIP 2 Vision Backbone

This system is built on SigLIP 2 [4]. We used the specific version from Hugging Face (`Ateeqq/ai-vs-human-image-detector`), which is already set up to tell the difference between real photos and AI-generated images. The text encoder and cross-modal projection components are surgically removed, leaving only the image encoder—a Vision Transformer (ViT) architecture with a patch size of 16 pixels and 1,024-dimensional patch embeddings. Retaining only the image encoder reduces the model's parameter count and memory footprint substantially while preserving the rich visual representations learned during the image-text alignment pre-training phase.

The position embeddings of the resulting vision encoder were originally pre-trained at a resolution of 224×224 pixels, corresponding to a 14×14 grid of 196 patches. To operate at 512×512 pixels, which preserves forensically important fine-grained details, the position embedding matrix is bicubically interpolated from the original 14×14 grid to a 32×32 grid of 1,024 patches. This interpolation is applied once before training and the new position embedding weights are treated as trainable parameters, allowing them to adapt to the forensic classification task during the subsequent fine-tuning phase. The image encoder thus accepts images of size $512 \times 512 \times 3$ and produces an output embedding of dimension 1,024.

4.4.2 Traditional Forensic Feature MLP

The seven-dimensional forensic feature vector extracted from each image is processed by a dedicated Feature MLP comprising three fully connected layers with the following structure:

- Input layer: $7 \to 64$ with LayerNorm and GELU activation
- Intermediate layer: $64 \to 128$ with LayerNorm and GELU activation
- Output layer: $128 \to 128$

Dropout regularisation is applied after each non-linear activation. This MLP learns to embed the raw forensic feature scores into a 128-dimensional learned representation, allowing the model to discover non-linear combinations and interactions among the seven features that are more discriminative than the individual scores alone.

4.4.3 Hybrid Classification Head with SE-Attention

The 1,024-dimensional backbone embedding and the 128-dimensional forensic feature embedding are concatenated to produce a 1,152-dimensional fused representation. This fused vector is then processed by the Hybrid Forensic Head, a three-layer classification network:

- **Layer 1:** Linear($1152 \to 1024$), BatchNorm1d, GELU activation, 40% Dropout.
- **Attention:** SE-style ArtifactAttention module — a squeeze-and-excitation network that computes a channel-wise attention weight vector from the 1,024-dimensional representation through a bottleneck MLP ($1024 \to 256 \to 1024$) with Sigmoid activation, then multiplies these weights element-wise into the representation.
- **Layer 2:** Linear($1024 \to 512$), BatchNorm1d, GELU activation, 30% Dropout.

- **Layer 3 (Output):** Linear($512 \rightarrow 2$), producing raw logits for the two classes.

The SE style attention mechanism is the architectural innovation that distinguishes this head from regular classifiers. The attention module performs an implicit feature selection operation that enhances the most informative features for AI artifact detection while down-weighting irrelevant or noisy features.

4.5 Training Strategy

The model was trained using a carefully designed strategy that accounts for the heterogeneous nature of the hybrid architecture, balancing the pre-trained backbone against the randomly initialised classification components. The following subsections detail the hardware setup, the freeze-unfreeze schedule, the loss formulation, and the complete training algorithm.

4.5.1 Training Setup

Training was performed on Kaggle Notebooks using $2\times$ T4 GPUs with 16 GB VRAM each. The model was trained for a maximum of 25 epochs with early stopping based on validation AUC, with a patience of 3 epochs. The effective batch size was 32 (8 per device $\times$ 2 GPUs $\times$ 2 gradient accumulation steps). Input images were resized to 512×512 during training using a smart adaptive resize strategy: images larger than 1024×1024 were first downsampled to 1024×1024 using LANCZOS anti-aliasing before the final bicubic resize to 512×512, to avoid aliasing artefacts in the forensic features. Training augmentations were restricted to forensically-safe operations: random JPEG compression (quality 70–98, applied with probability 0.4), quality balancing (probability 0.25), and random resized crop.

4.5.2 Backbone Freeze-Unfreeze Schedule

To prevent the rich SigLIP 2 representations from being immediately distorted by gradient updates from the randomly initialised classification head, the backbone was frozen during the first two training epochs. During this phase, only the Feature MLP and Hybrid Head parameters were updated, allowing the classification components to develop reasonable initial weights. From epoch 3 onwards, the backbone was unfrozen and all parameters were jointly optimised with a differential learning rate: a base learning rate of 2×10^{-5} for the Feature MLP and Head, and a $10\times$ reduced learning rate (2×10^{-6}) for the backbone parameters to prevent catastrophic forgetting.

4.5.3 Loss Function and Class Weighting

The training objective was class-weighted cross-entropy loss with label smoothing ($\varepsilon = 0.05$). Class weights were computed inversely proportional to class frequency in the training split: the AI class (3,200 images, 42.1%) received a weight of 1.19, and the real class (4,400 images, 57.9%) received a weight of 0.87. The loss function is defined as:

$$\mathcal{L}(\hat{y}, y) = -\sum_{c} w_c \cdot \left[(1-\varepsilon)y_c + \frac{\varepsilon}{K}\right] \cdot \log(\hat{y}_c) \tag{4.4}$$

where $K = 2$ classes, w_c is the class weight, and ε is the label smoothing factor. The AdamW optimiser was used with a cosine annealing learning rate schedule and a linear warmup phase covering the first 10% of training steps.

4.5.4 Formal Training Algorithm

Algorithm 4.3 presents the complete training pipeline.

Table 4.3: Algorithm 4.1: Hybrid AI-Generated Image Detector Forensic Training Pipeline.

Input: Image dataset $D = \{(x_i, y_i)\}$ where $x_i \in \mathbb{R}^{3\times512\times512}$, $y_i \in \{0 = \text{AI}, 1 = \text{Human}\}$; pre-computed Phase-1 features CSV with 7 forensic signals per image; fine-tuned SigLIP backbone. **Output:** Fine-tuned hybrid classifier f_θ; evaluation metrics {Accuracy, F1, Precision, Recall, AUC-ROC, FPR, FNR}
Step 1: Hardware Initialisation a. Detect available CUDA devices; abort if no GPU found. b. Set DEVICE = “cuda”; configure DataLoader with fp16=True. c. Effective batch size = BATCH_PER_GPU $\times$ N_{GPU} $\times$ GRAD_ACCUM = $8 \times N_{GPU} \times 4$.
Step 2: Dataset Loading & Class-Weight Computation a. Recursively scan AI folder (label=0) and Real folder (label=1). b. Compute inverse-frequency class weights: $w_{AI} = N/(2N_{AI})$; $w_{real} = N/(2N_{real})$. c. Stratified 3-way split: Train 70% | Val 15% | Test 15%; random seed = 42.
Step 3: Phase-1 Feature Loading & Scaling a. Load pre-computed features.csv; verify all 7 feature columns exist. b. Apply sign inversion where needed; replace NaN with 0.5; clip all values to $[0, 1]$. c. Fit StandardScaler on full feature matrix; store in path $\rightarrow$ vector lookup table.

(Algorithm 4.1 continued)
Step 4: Model Architecture a. Load SigLIP backbone; resize position embeddings from 14×14 to 32×32 via bicubic interpolation. b. Attach FeatureMLP: $7 \to 64 \to 128 \to 128$ with LayerNorm and GELU. c. Attach HybridForensicHead: $(d_b+128) \to 1024 \to$ SE-Attention $\to 512 \to 2$.
Step 5: Training Loop (epochs $e = 1 \ldots 25$) a. Freeze backbone for first 3 epochs; train only FeatureMLP and HybridHead. b. At epoch 3: unfreeze backbone for full joint fine-tuning. c. Forward: backbone $\to$ img_emb; FeatureMLP $\to$ feat_emb; concatenate $\to$ HybridHead $\to$ logits. d. Compute weighted cross-entropy with label smoothing $\varepsilon = 0.05$. e. Accumulate gradients for 4 steps; fp16 mixed precision. f. Save checkpoint if F1 improves; early stop if no improvement for 5 epochs.
Step 6: Evaluation a. Run inference on held-out test set (15%) with no augmentation. b. Compute: Accuracy, Precision, Recall, F1, AUC-ROC, FPR, FNR.

4.6 Inference Pipeline

The inference pipeline was written in Python with PyTorch and Gradio for web interface deployment on Hugging Face Spaces. For each input image, the pipeline performs the following steps:

1. **Image loading and validation:** Pillow loads the input image with truncated image handling enabled. The image is converted to RGB if not already in that format.

2. **Adaptive resizing:** The image is resized to 512×512 using the adaptive resize strategy described in Section 4.5.1, with step-down only when the image exceeds 1024×1024 pixels.

3. **Forensic feature extraction:** The seven forensic features are computed from the full-resolution resized image. The raw feature values are normalised using the StandardScaler statistics computed from the training set and stored in the model checkpoint.

4. **Backbone inference:** The resized image is processed by the SigLIP 2 image processor (normalisation to ImageNet statistics) and passed through the backbone. The 1,024-dimensional embedding is extracted from the hidden states of the final transformer layer.

5. **Fusion and classification:** The backbone embedding and the normalised forensic feature embedding are concatenated and passed through the Hybrid Forensic Head to produce class logits.

6. **Output:** The softmax probabilities over the two classes are returned, along with the individual forensic feature scores for explainability. The total end-to-end latency measured on CPU hardware at Hugging Face Spaces is 142.4 milliseconds.

4.7 Tools and Technologies

Table 4.4: Tools and technologies used in development.

Tool / Library	**Version**	**Purpose**
Python	3.10	Primary programming language
PyTorch	2.3	Neural network implementation and training
HuggingFace Transformers	4.41	SigLIP 2 model loading and processing
HuggingFace Datasets	2.19	Dataset management and streaming
NumPy / SciPy	1.26 / 1.13	Forensic feature computation
PyWavelets	1.6	Haar Wavelet Transform implementation
Pillow (PIL)	10.3	Image loading and preprocessing
Scikit-learn	1.5	StandardScaler, metrics computation
Gradio	4.x	Web inference interface (Hugging Face Spaces)
Kaggle Notebooks	N/A	Training environment (dual T4 GPUs)

Chapter 5

Results and Discussion

5.1 Primary Performance Metrics

The model was tested on a separate test set of 1,132 images and had an overall accuracy of 96.6%. For the class generated by AI, the precision achieved 98.9%, recall 95.3% and an F1-Score of 97.0%, which reflects good balance in minimising false-positives and negative. The AUC-ROC score is calculated for the plots of 0.992 further demonstrates near perfect class-separability at every classification threshold. Inference latency was measured to be 142.4 ms on CPU using a Hugging Face Spaces environment which is still decent enough when it comes to real-world deployment without GPU resources.

Table 5.1: Primary performance metrics on the held-out test set.

Metric	**Value**
Accuracy	96.6%
Precision (AI class)	98.9%
Recall (AI class)	95.3%
F1-Score (AI class)	97.0%
AUC-ROC	0.992
Inference Latency (CPU, HF Spaces)	142.4 ms

5.2 Confusion Matrix Analysis

As shown in Figure 5.1, out of 472 images created by humans, 465 images were classified correctly and only 7 images were mistakenly flagged as AI-generated (false positive rate is only 1.5%). Of the 660 AI-generated images, 629 were correctly identified while 31 of those images were misclassified as human because of the likely fact that the AI images closely mimicked human artistic styles. The significantly lower number of false positives than false negatives is symptomatic of the model's tendency to be conservative in labelling human content as AI generated, which is a desirable property in sensitive deployment contexts, in which misclassification of content genuinely produced by human writers has more severe consequences.

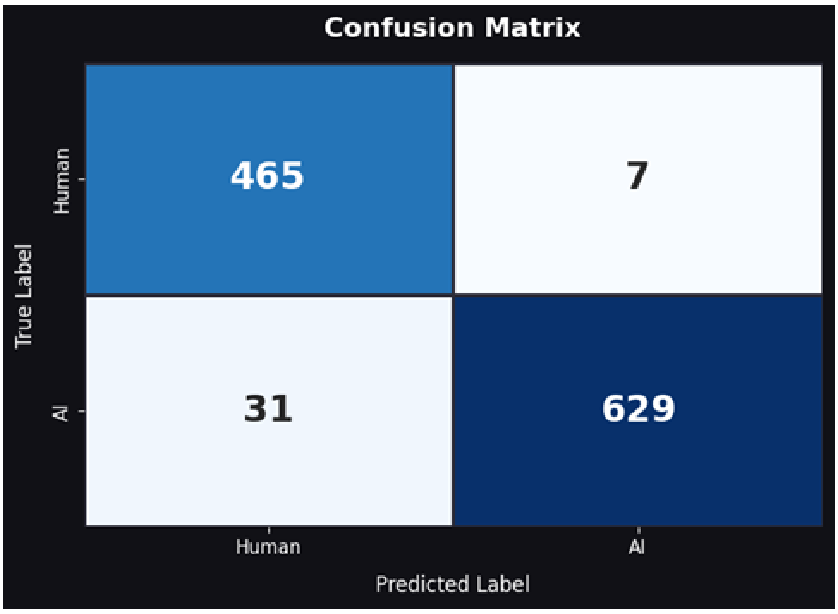

Figure 5.1: Confusion matrix on the held-out test set of 1,132 images.

5.3 ROC Curve and AUC Analysis

Receiver Operating Characteristic (ROC curve) is the probability curve for all possible TPR and FPR classification thresholds. An AUC-ROC of 0.992 means that the classifier

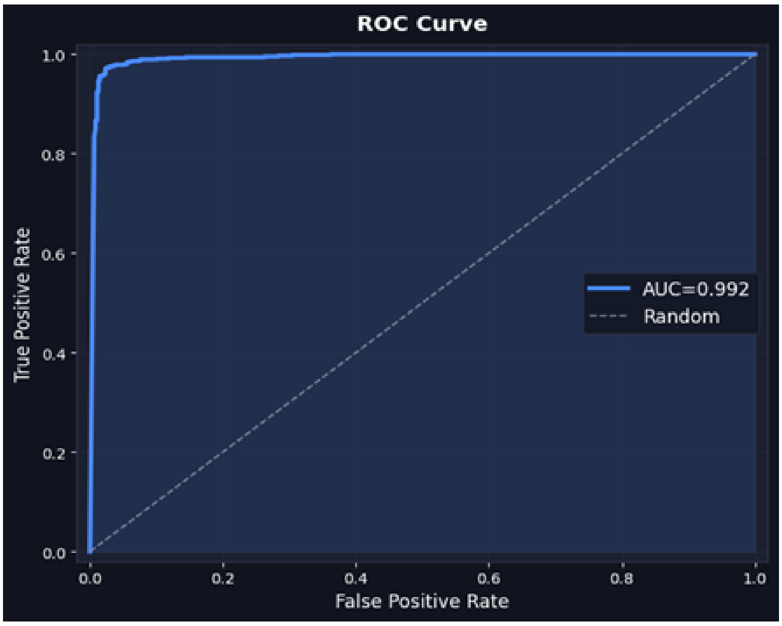

Figure 5.2: Receiver Operating Characteristic (ROC) curve for the proposed hybrid model.

can discriminate, or in laymen terms, there is a 99.2% of probability that if you randomly draw an AI “fake” image and a random actual image the fake image would get the higher probability score”. And from ROC curve - the model has its True Positive Rate (TPR) greater than 99% while still maintaining a False Positive Rate (FPR) lesser than 3%. “This clearly shows that a precision-recall trade off can be improved by lowering the classification threshold, without sacrificing recall”.

5.4 Comparison with Commercial Detection Tools

A qualitative comparative study was carried out using a set of hard-negative AI-generated images. These images are of high quality, with no obvious visual artifacts, making them challenging to detection models. The same set of images was passed through two detection tools, Undetectable AI and AI or Not, along with the proposed model, and the results obtained are shown in Table 5.2.

Table 5.2: Comparative analysis on a representative hard-negative AI-generated image.

Detector	**Verdict**	**Latency**
Proposed Model (Ours)	AI-Generated (100% confidence)	142.4 ms
AI or Not (Commercial)	Likely REAL	920 ms
Undetectable AI (Commercial)	98% REAL	5,230 ms

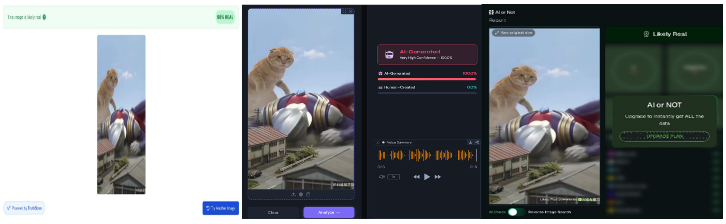

Figure 5.3: Hard-negative test case: comparison of detection results across our model and commercial tools.

The results demonstrate two important advantages of the proposed architecture. First, the proposed model correctly identifies the hard-negative image as AI-generated with 100% confidence, while both commercial tools misclassify it as real. This superior accuracy on hard negatives is attributable to the combination of the SigLIP 2 backbone's rich visual representations with the traditional forensic features, which collectively detect signatures that are not visually apparent but are statistically distinguishable. Second, the proposed model achieves this result in 142.4 milliseconds—6.5× faster than AI or Not (920 ms) and 36.7× faster than Undetectable AI (5,230 ms). At 142.4 ms per image, the proposed model can process approximately 7 images per second, making it viable for integration into real-time content moderation pipelines, whereas the commercial tools would process fewer than 1 image per second on the same hardware.

5.5 Discussion and Interpretation

The results demonstrate that the proposed hybrid architecture successfully achieves the primary research objective: a single-pass AI image detector that matches the accuracy of heavy ensemble systems while operating at inference speeds suitable for real-time deployment. The 0.992 AUC-ROC places the model firmly within the range of top-performing ensemble systems reported in prior benchmark studies [2, 12], while the 142.4 ms inference latency is a significant improvement over commercial platforms and approaches the real-time threshold.

The main conclusion supported by these findings is that traditional forensic features and deep visual features are complementary, rather than redundant, features. The SigLIP 2 backbone, trained on 10 billion image-text pairs [4], has learned features that are sensitive to high-level visual and semantic properties that distinguish AI-generated images from real images. The traditional features, obtained from physical models of camera image formation, capture low-level statistical properties in the frequency, noise, and texture domains, which are largely inaccessible to attention-based models operating at the feature map level. The use of the SE-style attention mechanism in the classification head learns how to dynamically weight the relative importance of the two signal streams, effectively performing a learned feature selection depending on the properties of each image artifact.

The improved performance on hard-negative images, especially those images classified correctly when commercial tools fail, provides perhaps the most compelling evidence supporting this hybrid approach. By definition, hard-negative images do not contain visually obvious AI artifacts, and detection relies on reasoning about statistical properties, which are physically meaningful and perceptually subtle. The traditional features, especially the Bayer noise pattern, which detects the absence of camera demosaicing fingerprints, and the DCT Benford deviation, which detects violations of natural frequency statistics, provide a discriminative signal that is absent in AI-generated images.

Chapter 6

Summary and Future Work

6.1 Summary of Contributions

This dissertation details the development and evaluation of a lightweight hybrid AI image detector. The system is built to balance detection accuracy with computational efficiency.The project introduces two main technical contributions. First, it adapts the SigLIP 2 vision encoder [4] for binary AI image forensics. While originally trained on 10 billion image-text pairs, this encoder was modified by removing the text components and applying bicubic position embedding interpolation to process 512×512 resolution inputs. This method keeps the foundation model's core visual representations intact but requires much less memory and computing power.Second, the system includes a forensic feature engineering module that calculates seven specific signals: LBP entropy, DCT Benford deviation, Gradient Co-occurrence entropy, Haar Wavelet HH energy, FFT power slope, Bayer noise inter-channel correlation, and Edge Sharpness grid. A Phase-1 selection study verified these metrics. Together, these computed signals capture the frequency, noise, and texture patterns that standard deep learning models usually miss.

A custom three-layer classification head incorporating SE-style channel attention was designed to fuse the 1,024-dimensional SigLIP 2 deep embedding with the 128-dimensional forensic feature embedding. The SE-attention mechanism provides a learned, dynamic weighting of feature dimensions that adapts to each input image.

The system was trained on a curated dataset of 7,600 images spanning five distinct AI generative paradigms and diverse real-world photographic sources, and achieved 96.6% accuracy, 98.9% precision, 95.3% recall, 97.0% F1-score, and 0.992 AUC-ROC on a held-out test set of 1,132 images—metrics that place it firmly within the accuracy range of best-in-class ensemble systems.

The system was deployed as an interactive Hugging Face Spaces application with a measured end-to-end inference latency of 142.4 milliseconds on CPU hardware, representing a $6.5\times$ and $36.7\times$ speed advantage over the commercial tools AI or Not and Undetectable AI respectively. On a representative hard-negative test case, the proposed model correctly identified a synthetic image that both commercial tools classified as real, with 100% confidence.

6.2 Concluding Remarks

The main hypothesis of this dissertation, that is, the possibility of achieving the accuracy of intensive ensemble detection systems with significantly less computational cost using a one-pass hybrid architecture with a SigLIP2 vision backbone and deliberately designed traditional forensic features, has been supported by the experiments. The accuracy of 96.6%, AUC-ROC of 0.992, and inference latency of 142.4 ms, along with better performance on hard-negative images compared to commercial tools, demonstrate that the long-standing efficiency-accuracy trade-off in AI-based image detection systems is, in fact, not an intrinsic property. This trade-off is an artifact that can be overcome with deliberate architecture design, capitalizing on the strengths of both deep visual representations and traditional forensic features.

Generative AI-based imagery is one of the most complex and challenging problems to the integrity of digital information in recent times. The tools and architectures developed in this dissertation are an important step towards developing accessible, accurate, and real-time solutions to this problem. However, they are just an initial step towards resolving the issue. The generative AI-based tools will continue to evolve to counter the detection tools developed in this dissertation. Moreover, the forensic features used in these detection tools will continue to degrade. A more effective solution to this problem will require more than just better detection tools. A more effective solution will require changes to the image generation and distribution ecosystem. A more effective solution will require better image provenance standards, the use of watermarking in image generation tools, and policy changes to assign responsibility to image generators.

6.3 Limitations

Although the quantitative outcomes are strong, several limitations should be acknowledged. The 7,600 diverse training images are small compared to detectors trained on state-of-the-art datasets such as UnivFD or GenImage (100,000+ images). This dataset size likely limits the model's ability to generalise to generative models and photographic styles not represented in training.

The forensic feature extraction step is rapid (less than 10 ms of the total inference time) but relies on signal processing libraries (SciPy, PyWavelets) that may present challenges in constrained deployment environments. The Bayer noise feature, while theoretically powerful, can give unreliable indicators for images that have undergone substantial JPEG re-compression, or for images from cameras with non-standard demosaicing algorithms.

The model was evaluated on a single held-out test split drawn from the same distribution as the training data. Although distribution shift is acknowledged as the primary challenge for all existing detectors—as demonstrated by the AIGIBench benchmark studies [2] and the zero-shot evaluation by Ren et al. [3]—the current work lacks a formal cross-dataset generalisation evaluation on external benchmarks.

6.4 Future Work

There are several promising directions for future research following from the limitations and findings of this work:

- **Larger and more diverse training datasets:** Increasing the size of the training dataset beyond 50,000 images is also likely to improve accuracy, both for known and unknown data sets. This would include generative architectures developed after the initial dataset collection. To address the problem of evolving image generation technologies, the evaluation framework itself needs to be tested with updated benchmarks such as AIGIBench and GenImage.

- **Cross-dataset generalisation evaluation:** An evaluation of the proposed model's generalisation potential using external test datasets (e.g., ForenSynths, GenImage, and WildFake) would help to clearly identify the model's generalization performance and enable comparison with previously proposed models, wherein the generalization performance is mentioned.

- **Adversarial robustness:** The performance of the proposed model in the context of adversarial attacks, wherein an adversary tries to attack an AI-based image detection system by introducing minor pixel-level changes, is unknown. The proposed model's performance against image distortions, as mentioned in the AIGIBench test protocol, is yet to be determined. The image distortions include JPEG compression, image resizing, Gaussian noise, and color jittering.

- **Explainability and visual attribution:** Although the proposed model provides the facility to make the model explainable through the determination of forensic feature scores, the image regions that contributed to the classification are unknown. The determination of the model's potential to integrate the explainability techniques with the proposed model, such as the SigLIP 2 model, and generating heat maps to provide tangible image evidence to support the human evaluators' trust in the model's classification results is yet to be determined.

- **Video and multi-frame detection:** In this paper we deal with images one by one, an extension of this framework to video data would be a natural development. Especially since unusual inter-frame noise correlation and optical flow features could act as discriminative features. This is particularly pressing with the growing popularity of AI-generated video content on Sora.

- **Model compression and edge deployment:** Even though the SigLIP 2 model is much smaller than other vision and language models, it still consumes a few hundred megabytes of memory. Structured pruning, knowledge distillation, quantization (INT8 or INT4) etc. are likely to compress the model size even further and considerably reduce latency making it easy to deploy on edge devices without requiring GPU acceleration.

References

[1] Bilva Chandra. Analyzing harms from AI-generated images and safeguarding online authenticity. Technical report, RAND Corporation, 2024. URL `https://doi.org/10.7249/PEA3131-1`.

[2] Ziqiang Li, Jiazhen Yan, Ziwen He, Kai Zeng, Weiwei Jiang, Lizhi Xiong, and Zhangjie Fu. Is artificial intelligence generated image detection a solved problem? *arXiv preprint*, 2025. doi: 10.48550/arXiv.2505.12335.

[3] Simiao Ren, Yuchen Zhou, Xingyu Shen, Kidus Zewde, Tommy Duong, George Huang, Hatsanai Tiangratanakul, Tsang Ng, En Wei, and Jiayu Xue. How well are open sourced AI-generated image detection models out-of-the-box: A comprehensive benchmark study. *arXiv preprint*, 2026. doi: 10.48550/arXiv.2602.07814.

[4] Michael Tschannen, Alexey Gritsenko, Xiao Wang, Muhammad Ferjad Naeem, Ibrahim Alabdulmohsin, Nikhil Parthasarathy, Talfan Evans, Lucas Beyer, Xiaohua Zhai, and Alexander Kolesnikov. SigLIP 2: Multilingual vision-language encoders with improved semantic understanding, localisation, and dense features. *arXiv preprint*, 2025. doi: 10.48550/arXiv.2502.14786.

[5] Ian Goodfellow, Yoshua Bengio, and Aaron Courville. *Deep Learning*. MIT Press, 2016.

[6] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision*

and Pattern Recognition (CVPR), pages 770–778, 2016. doi: 10.1109/CVPR.2016.90.

[7] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, et al. An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint*, 2020. doi: 10.48550/arXiv.2010.11929.

[8] NaHyeon Park, Kunhee Kim, Junsuk Choe, and Hyunjung Shim. Rethinking the use of vision transformers for AI-generated image detection. *arXiv preprint*, 2025. doi: 10.48550/arXiv.2512.04969.

[9] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. Learning transferable visual models from natural language supervision. In *Proceedings of the International Conference on Machine Learning (ICML)*, 2021. doi: 10.48550/arXiv.2103.00020.

[10] Huan Liu, Zichang Tan, Chuangchuang Tan, Yunchao Wei, Yao Zhao, and Jingdong Wang. Forgery-aware adaptive transformer for generalizable synthetic image detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. doi: 10.48550/arXiv.2312.16649.

[11] Nicolò Bonettini, Paolo Bestagini, Sara Mandelli, and Stefano Tubaro. On the use of Benford's law to detect GAN-generated images. In *Proceedings of the International Conference on Pattern Recognition (ICPR)*. IEEE, 2020. doi: 10.48550/arXiv.2004.07682.

[12] Alan Perotti, Marco Nurisso, and Mirko Zaffaroni. No detector to rule them all. In *Proceedings of the 1st Deepfake Forensics Workshop (DFF'25) at ACM Multimedia 2025*, 2025. doi: 10.1145/3746265.3759659.

[13] Dimitrios Karageorgiou, Symeon Papadopoulos, Ioannis Kompatsiaris, and Efstratios Gavves. Any-resolution AI-generated image detection by spectral learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. doi: 10.48550/arXiv.2411.19417.

[14] Nan Zhong, Yiran Xu, Sheng Li, Zhenxing Qian, and Xinpeng Zhang. PatchCraft: Exploring texture patch for efficient AI-generated image detection. *arXiv preprint*, 2024. doi: 10.48550/arXiv.2311.12397.

[15] Adrian Lokner Lađević, Renata Kramberger, et al. Detection of AI-generated synthetic images with a lightweight CNN. *AI*, 5(3), 2023. doi: 10.3390/ai5030076.

[16] Pranav Gupta and Larry McMahan. Optimizing AI-generated image detection using a convolutional neural network model with fast fourier transform. *Journal of Emerging Investigators*, 2025. doi: 10.59720/24-270.

[17] Geeta Rani, Atharv Kothekar, Shawn George Philip, Vijaypal Singh Dhaka, Ester Zumpano, and Eugenio Vocaturo. Lightweight and hybrid transformer-based solution for quick and reliable deepfake detection. *Frontiers in Big Data*, 2025. doi: 10.3389/fdata.2025.1521653.

[18] Harshit, Y.V.S. Sarath, et al. Deepfake detection using EfficientNet-B0 and GRU. *International Journal of Engineering Research & Technology (IJERT)*, 2025. doi: 10.5281/zenodo.18106890.

[19] Chuangchuang Tan, Yao Zhao, Shikui Wei, Guanghua Gu, and Yunchao Wei. Learning on gradients: Generalized artifacts representation for GAN-generated images detection. In *2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 12105–12114, 2023. doi: 10.1109/CVPR52729.2023.01165.

[20] Timo Ojala, Matti Pietikäinen, and Topi Mäenpää. Multiresolution gray-scale and rotation invariant texture classification with local binary patterns. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 24(7):971–987, 2002. doi: 10.1109/TPAMI.2002.1017623.

Appendix A

Source Code and Implementation Details

A.1 Training Pipeline

The complete training pipeline is implemented in Python using PyTorch. The primary training script (`train.py`) handles dataset loading, model initialisation, the freeze-unfreeze training schedule, and checkpoint management as described in Algorithm 4.1 in Chapter 4.

A.2 Forensic Feature Extraction Module

The forensic feature extraction module (`features.py`) implements all seven forensic feature extractors described in Chapter 4. Each extractor accepts a PIL Image and returns a scalar in $[0, 1]$:

- `compute_lbp_entropy(image)` — Local Binary Pattern entropy
- `compute_dct_blocking(image)` — DCT Benford's Law deviation

- `compute_gradient_cooccurrence(image)` — Gradient co-occurrence entropy
- `compute_haar_wavelet(image)` — Haar wavelet diagonal energy ratio
- `compute_fft_slope(image)` — FFT power spectrum slope deviation
- `compute_bayer_noise(image)` — Bayer noise inter-channel correlation
- `compute_edge_sharpness(image)` — Edge sharpness grid score

A.3 Deployment Application

The Gradio web application (`app.py`) is deployed on Hugging Face Spaces and is publicly accessible. The application provides an interactive interface for uploading images and receiving real-time AI/real classification verdicts with confidence scores and individual forensic feature breakdowns.

```
Model ID: Ateeqq/ai-vs-human-image-detector
Deployment: Hugging Face Spaces
Inference Hardware: CPU (2 vCPU, 16 GB RAM)
Average Latency: 142.4 ms per image
```

A.4 Repository Structure

The complete source code repository is organised as follows:

```
hybrid-ai-detector/
|-- train.py            # Main training pipeline (Algorithm 4.1)
|-- features.py         # Forensic feature extraction module
|-- model.py            # HybridAIDetector architecture
|-- dataset.py          # Dataset loading and preprocessing
```

```
|-- app.py               # Gradio inference application
|-- requirements.txt     # Python dependencies
|-- backbone/            # Saved SigLIP 2 backbone weights
|-- hybrid_head.pt       # FeatureMLP + HybridForensicHead weights
|-- hybrid_config.json   # Feature metadata and config
```